

# AKHIL PRAGALLAPATI

## Data Engineer

New York (NY) • (607)-4541240 • akhilsai1603@gmail.com • linkedin.com/in/akhil-pragallapati/

## SUMMARY

---

Data Engineer with 3+ years of experience designing and optimizing data infrastructures for storage, processing, and analytics. Proficient in building ETL/streaming pipelines using Hadoop, Spark, Kafka, and Airflow, processing large datasets to drive data-driven decisions. Experienced in big data tools (Hive, Pig, HBase, MapReduce) and cloud technologies (AWS, Snowflake). Skilled in database management (MongoDB, PostgreSQL, Oracle) and ETL tools (Informatica, Databricks, Azure Data Factory). Advanced in data visualization (Tableau, Power BI) and Python for data analysis and machine learning.

## PROFESSIONAL EXPERIENCE

---

### Data Engineer

May 2021 – Jul

2022

Tata Consultancy Services, India

- Automated 50+ ETL process, reducing manual intervention by 200 hours annually and accelerating data processing for 10+ workflows.
- Incorporated SQL Server Integration Services (SSIS) for ETL, streamlining data extraction, transformation, and loading, reducing processing errors from 15 to 5 and improving load times.
- Administered 5+ MongoDB instances, executing 100+ complex queries, halving database response times from 100ms to 50ms, ensuring optimal performance.
- Implemented Waterfall project management for 10+ data engineering projects, ensuring structured progress and on-time delivery.

### Junior Data Engineer

Aug 2020 – May

2021

Tata Consultancy Services, India

- Created 10+ advanced Seaborn visualizations, converting complex data into actionable insights, aiding decisionmaking for 5+ teams.
- Deployed 10+ Power BI dashboards, integrating data from 5+ sources, saving 15 hours per month in analysis time.
- Applied data quality checks and ETL workflows with Informatica PowerCenter, reducing data inconsistencies from 10 to 7 and boosting operational efficiency.
- Designed a PostgreSQL database model on AWS RDS for a global client, managing 5+ terabytes of data and reducing query response times from 10 seconds to 7.5 seconds.

### Technical Data Analyst

Jun 2019 – Jul

2020

Vedanta Group, India

- Developed 10+ Python scripts for data manipulation and cleansing using libraries like NumPy and Pandas to analyze machinery performance data, improving data accuracy for operational insights across 5+ teams.
- Configured and managed 4 deployment pipelines (PROD, UAT, DEV, TEST) and 17 virtual environments using Azure DevOps, streamlining deployment processes and improving system reliability across 3 production environments.
- Enhanced 10+ Tableau reports with interactive features, enabling 20+ stakeholders to analyze machinery and production data, improving decision-making.
- Leveraged AWS EC2 for scalable data processing, optimizing resource utilization across 5+ data processing tasks and improving processing efficiency, ensuring high performance under varying workloads.

## EDUCATION

---

## TECHNICAL SKILLS

---

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| <b>Programming Languages:</b>    | Python, SQL                                                                |
| <b>Reporting Tools:</b>          | Tableau, PowerBI, MS Excel                                                 |
| <b>Python Libraries:</b>         | NumPy, Pandas, Scikit-learn, Scipy, Matplotlib, Seaborn                    |
| <b>Database:</b>                 | MongoDB, PostgreSQL, Oracle, Snowflake, IBM DB2, MySQL                     |
| <b>Big Data Tools:</b>           | Map Reduce, Spark, Airflow, HBase, Hive, Pig, Kafka, Hadoop                |
| <b>Database Management:</b>      | MySQL, PostgreSQL, MongoDB                                                 |
| <b>Cloud Technologies:</b>       | AWS (EC2, S3, RDS, IAM, CloudWatch, Lambda)                                |
| <b>Tools &amp; Technologies:</b> | Git, Docker, Jenkins, Docker, Kubernetes, Terraform                        |
| <b>ETL Tools:</b>                | Informatica PowerCenter, Informatica Cloud, Databricks, Azure Data Factory |
| <b>Methodologies:</b>            | Agile, Waterfall                                                           |

## PROJECTS

---

### Binghamton University, CRM Application

- Engineered a CRM web application leveraging the MEAN stack (MongoDB, Express.js, Angular, Node.js) with robust CRUD operations.
- Streamlined testing and deployment using CI/CD pipelines. Authored comprehensive documentation for architecture, APIs, and deployment workflows.

### Binghamton University, Real-Time Scalable Data Architecture Implementation

- Built a scalable data solution on Snowflake, increasing processing capacity to handle 700 million records daily (from 500 million).
- Established real-time data pipelines with Spark Streaming and Kafka, achieving a latency of less than 5 seconds for data ingestion.

### Binghamton University, Research Article Search Engine

- Integrated PostgreSQL as the primary database management system for handling backend operations, ensuring efficient data storage, retrieval, and management.
- Constructed the front-end using Django REST framework for API development, HTML, CSS, and Bootstrap for a responsive user interface, and JavaScript for interactivity.

### SASTRA University, Research Paper

- Analyzed a load balancing system specifically for stroke patients, optimizing resource allocation and treatment processes, which resulted in a 20% reduction in recovery time.